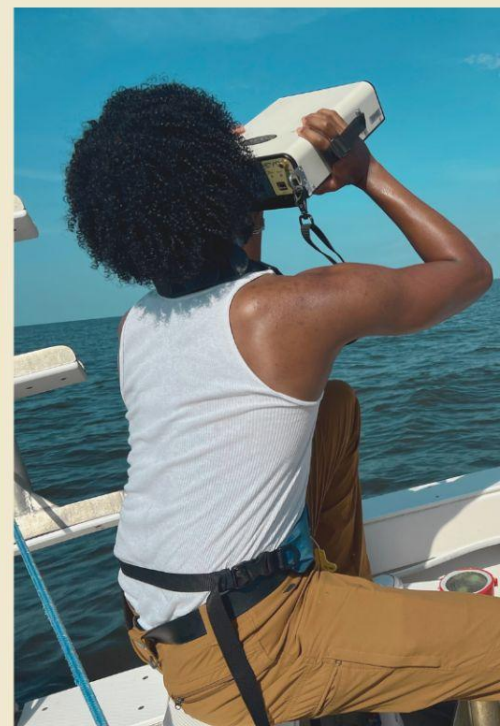
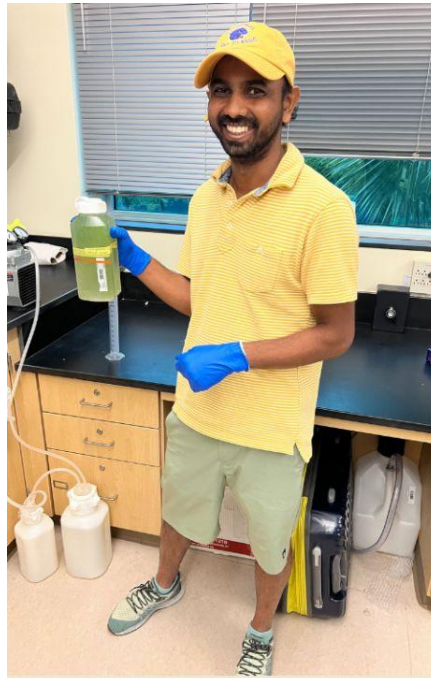


Introduction to Machine Learning in Aquatic Remote Sensing

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Department of Geography
University of Georgia
Athens, GA, USA**

Credits

- My lab mates
- PACE Hackweek 2025
- NASA Cryocloud
- INCOIS
(Dr. N.S. Rao)



Outline

- Part-1
 - Interactive Ice-breaker, Recap of Aquatic RS, Why ML?
- Part-2
 - Broad overview of standard ML/DL functioning , Scientific ML
- Part-3
 - Ethics & Metrics – R2 and beyond, Log-scale, visualizations
- Part-4
 - Tools for implementation, why Python is dominating, ethical use of LLMs

Let's get started!

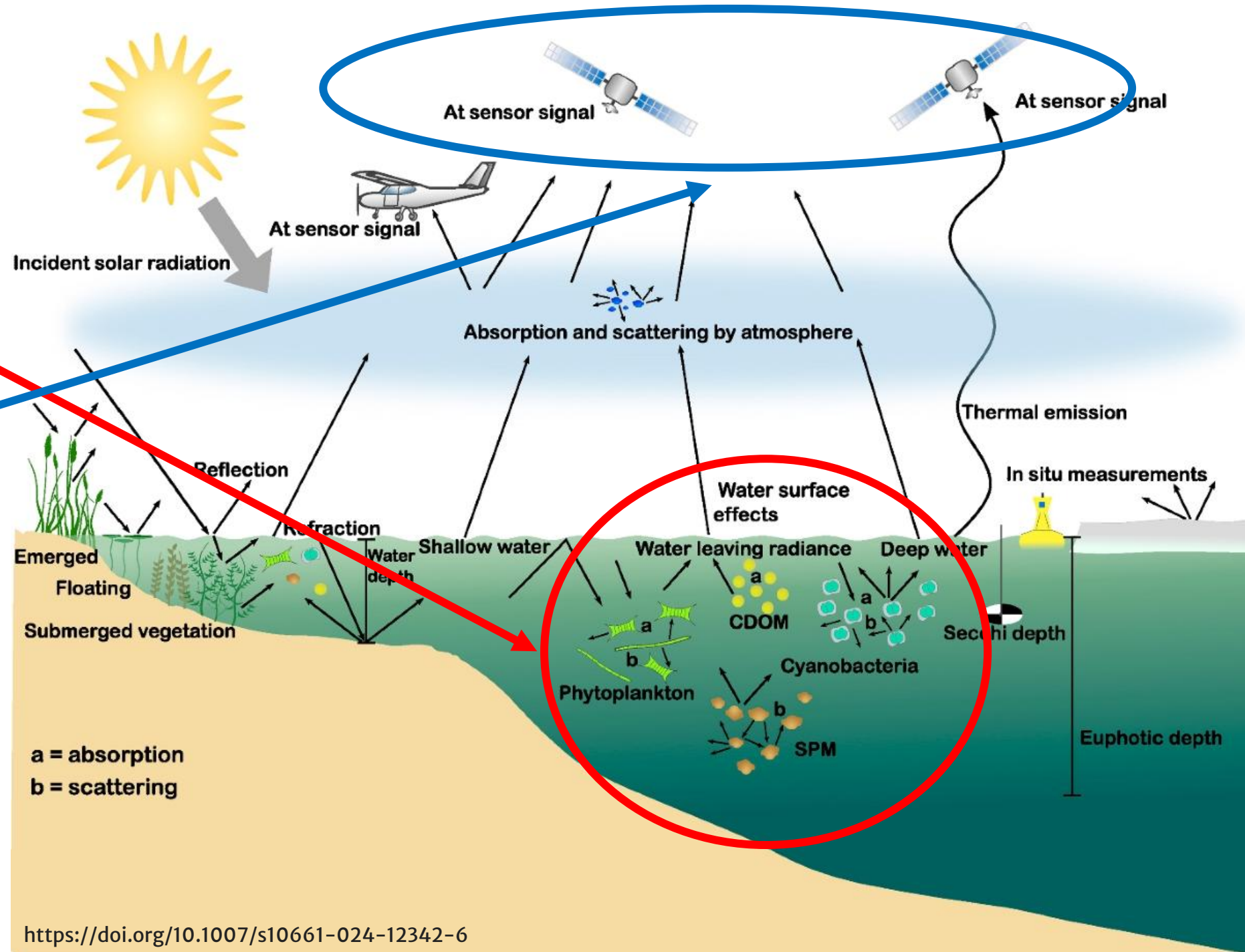
- **Please go to: menti.com**
- **Use code: 4379 7160**



Mentimeter results

What is Aquatic/Ocean
Color Remote Sensing?

- IOPs: Cause
- AOPs: Effect

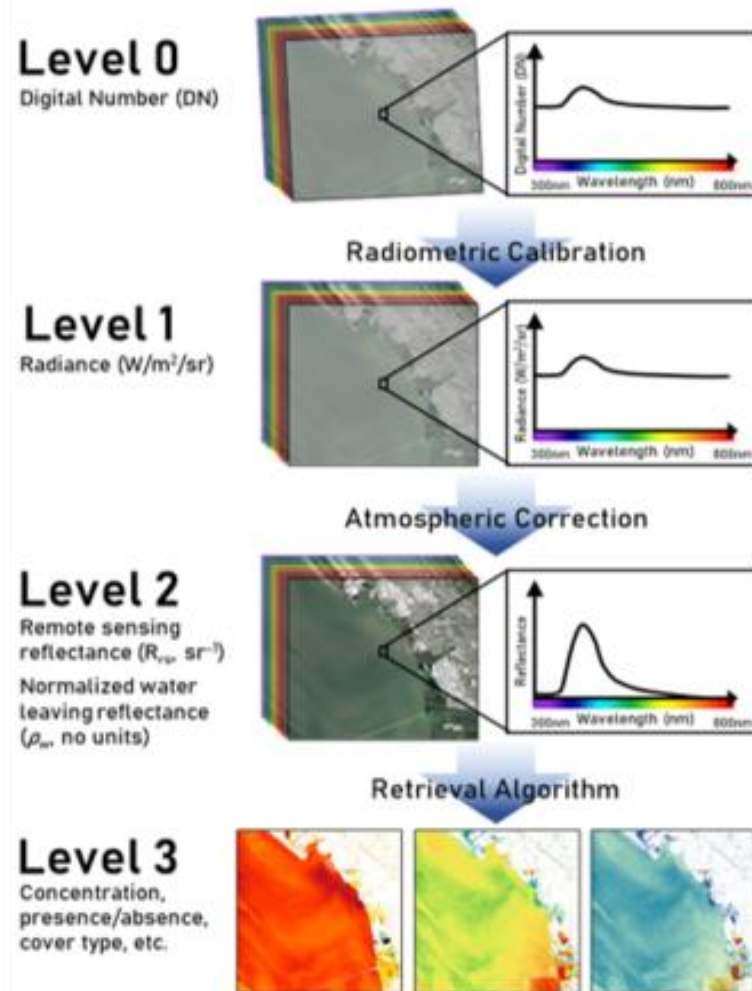


Light's interaction with water: What satellite sees

AOP

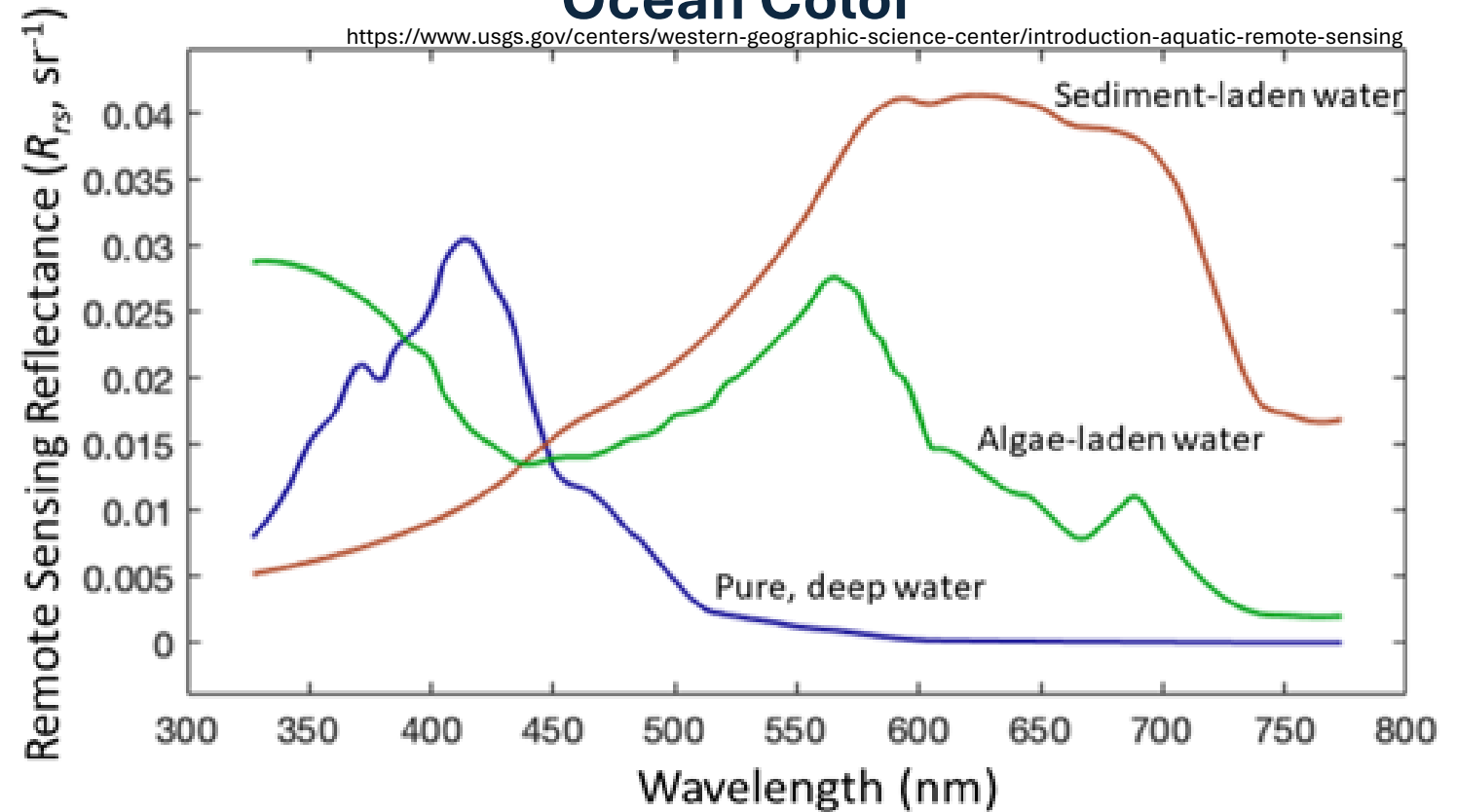
I
N
V
E
R
S
E

IOP



Ocean Color

<https://www.usgs.gov/centers/western-geographic-science-center/introduction-aquatic-remote-sensing>



Modelling Aquatic Processes

- Forward: IOPs (a,b) $\rightarrow$ [Forward Model] $\rightarrow$ AOPs [R_{rs}]
 - Unique
 - Natural
- Inverse: AOPs [R_{rs}] $\rightarrow$ [?] $\rightarrow$ IOPs (a,b)
 - Non-unique
 - Same spectra can represent multiple IOPs due to variation in sun-sensor geometry
- As remote sensing scientists, what do we do?
 - Inverse modeling

Why aquatic remote sensing? (NOAA, STAR - 2025)

- *Protect, restore and manage the use of coastal and ocean resources through ecosystem-based management*
- *Serve society's needs for weather and water information*
- *How to provide accurate water optical, biological, and biogeochemical property data in coastal and inland regions from satellite measurements?*

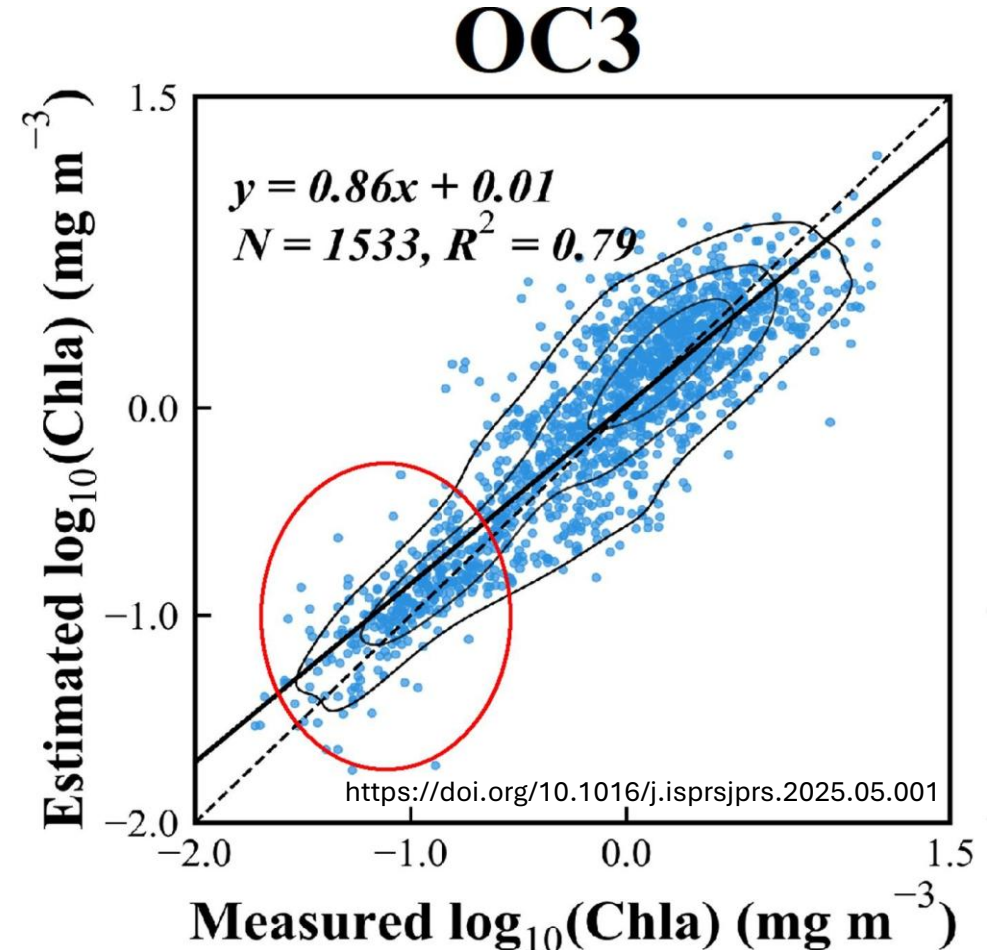
Benefit

- Protect and monitor our ocean resource
- Improve water resources forecasting capabilities
- Protect and monitor water resources
- Understand the effect of environmental factors on human health and well-being

Methods of Inverse Modeling

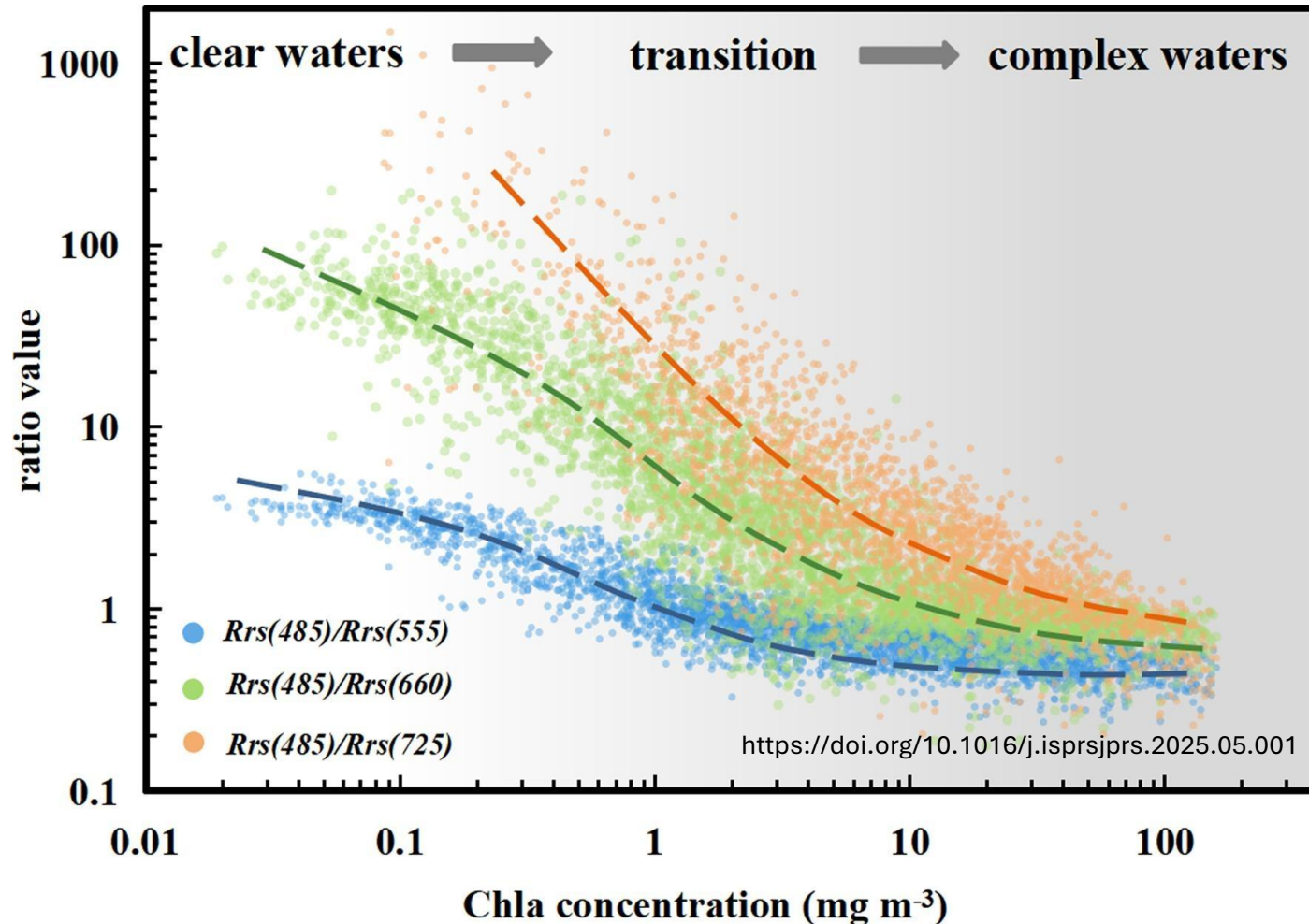
AOPs [R_{rs}] $\rightarrow$ [?] $\rightarrow$ IOPs (a,b)

- Band ratios, line heights
- Empirical and semi-empirical algorithms
- For instance, Chl-*a*:
 - OC4, .. , OCx
 - Blue-green ratio
 - NDCI
 - MCI
 - MPH



Why do we need ML then?

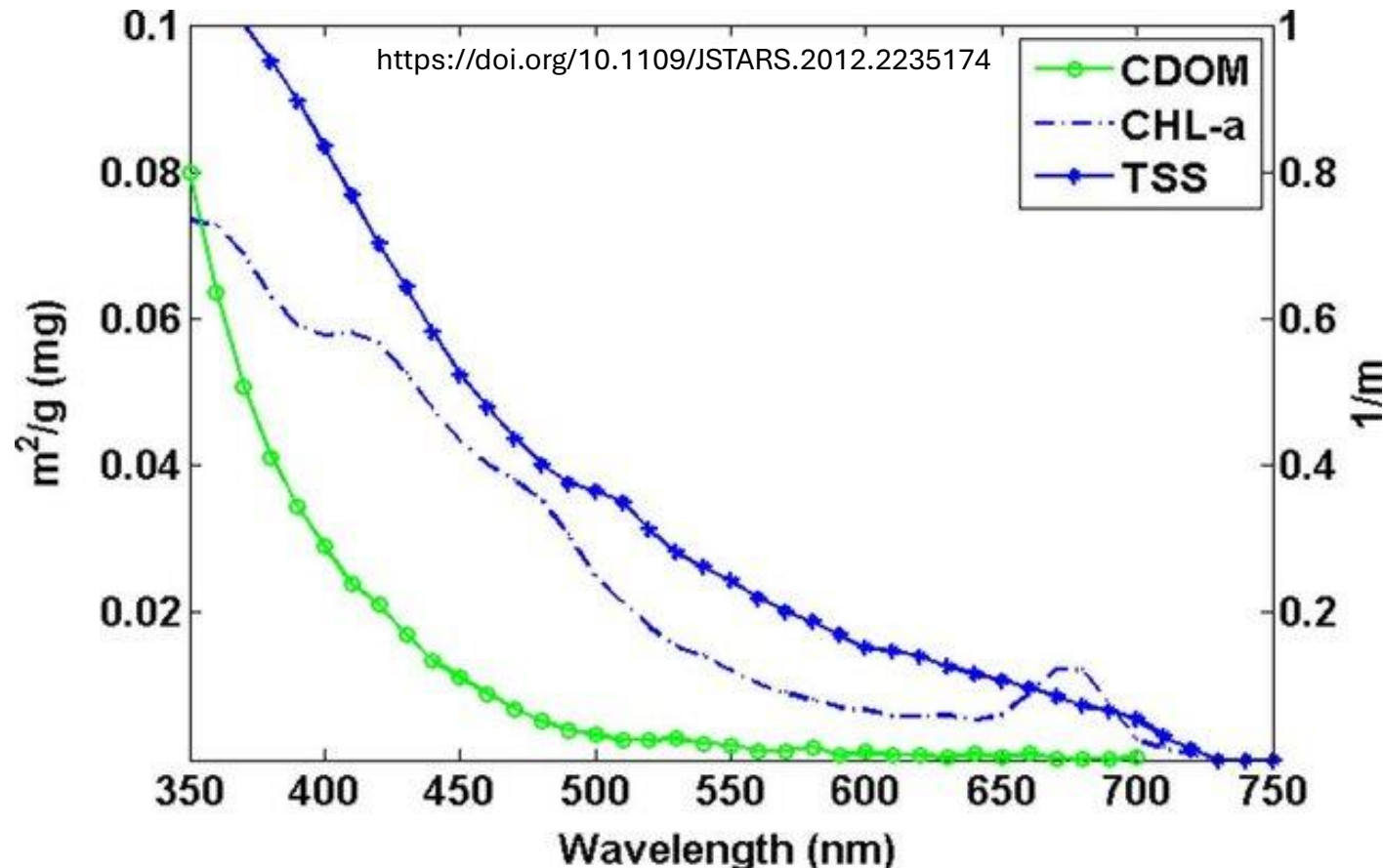
AOPs [R_{rs}] $\rightarrow$ [? ML ?] $\rightarrow$ IOPs (a,b)



- The empirical relationships hold theoretically
- With optically complex waters however, linear and apparent relationships stop holding

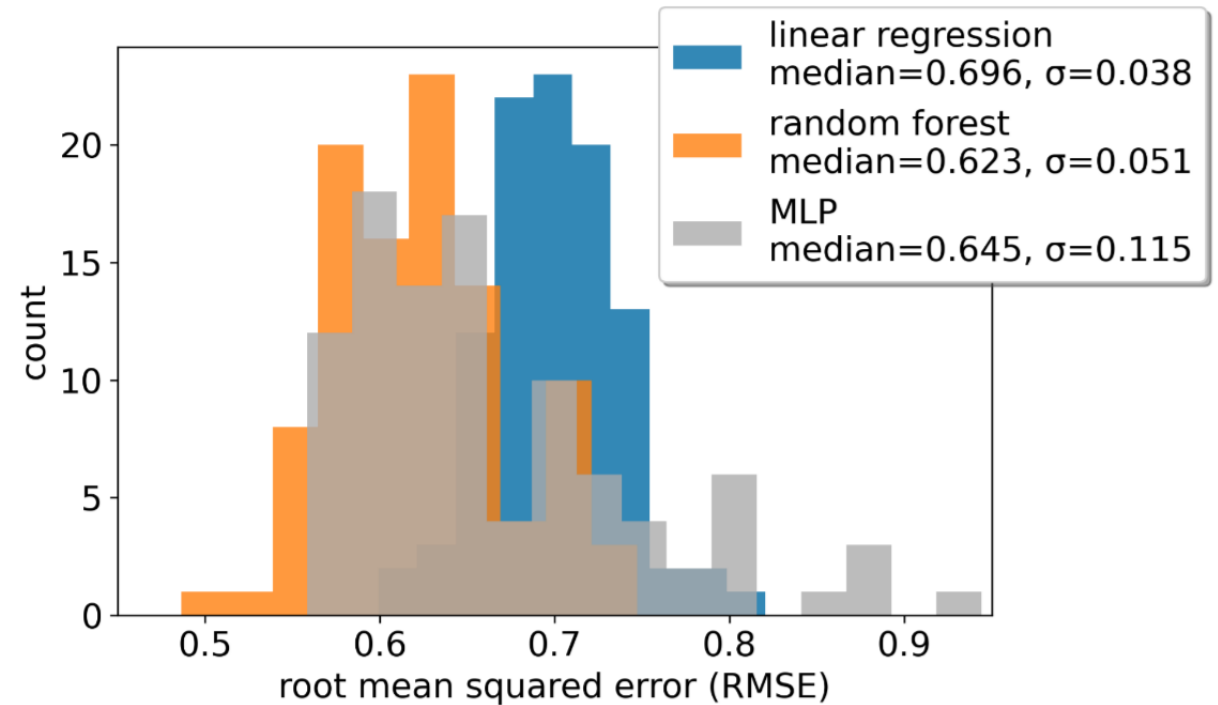
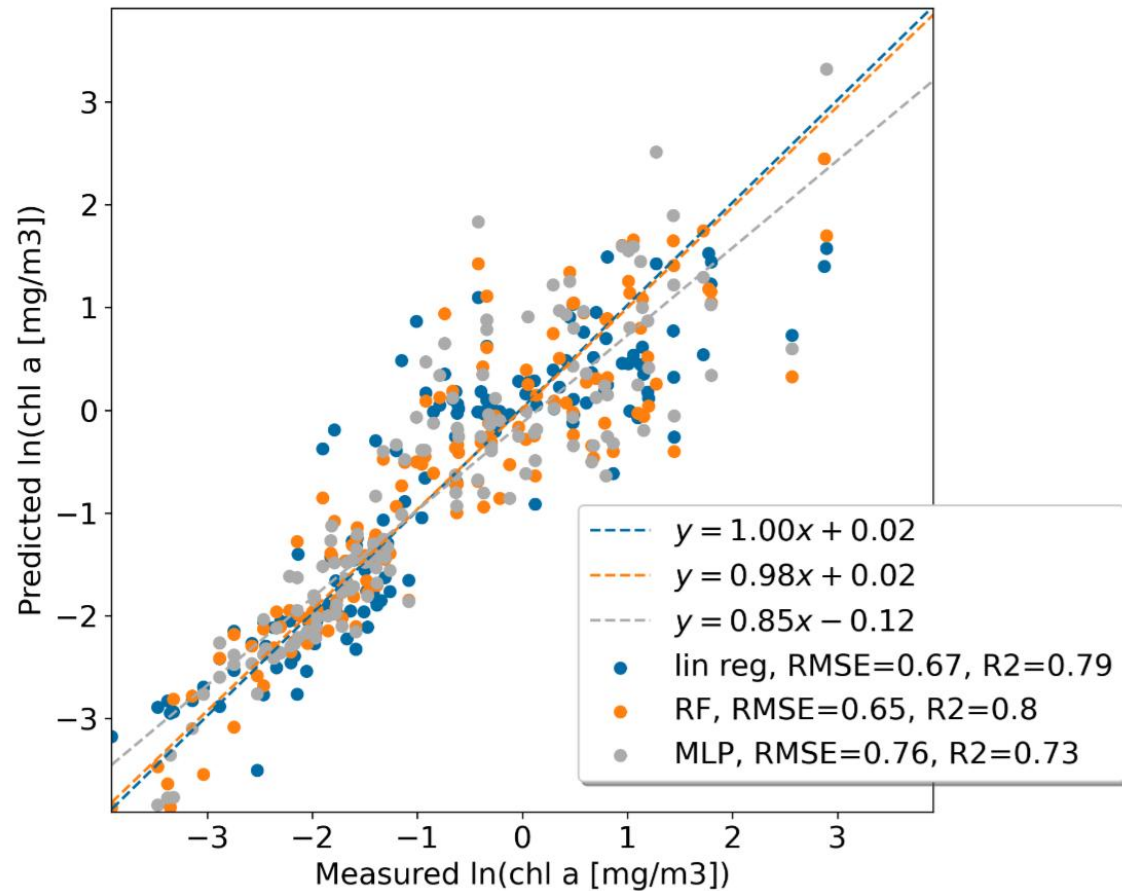
Why do we need ML then?

AOPs [R_{rs}] $\rightarrow$ [? ML ?] $\rightarrow$ IOPs (a,b)



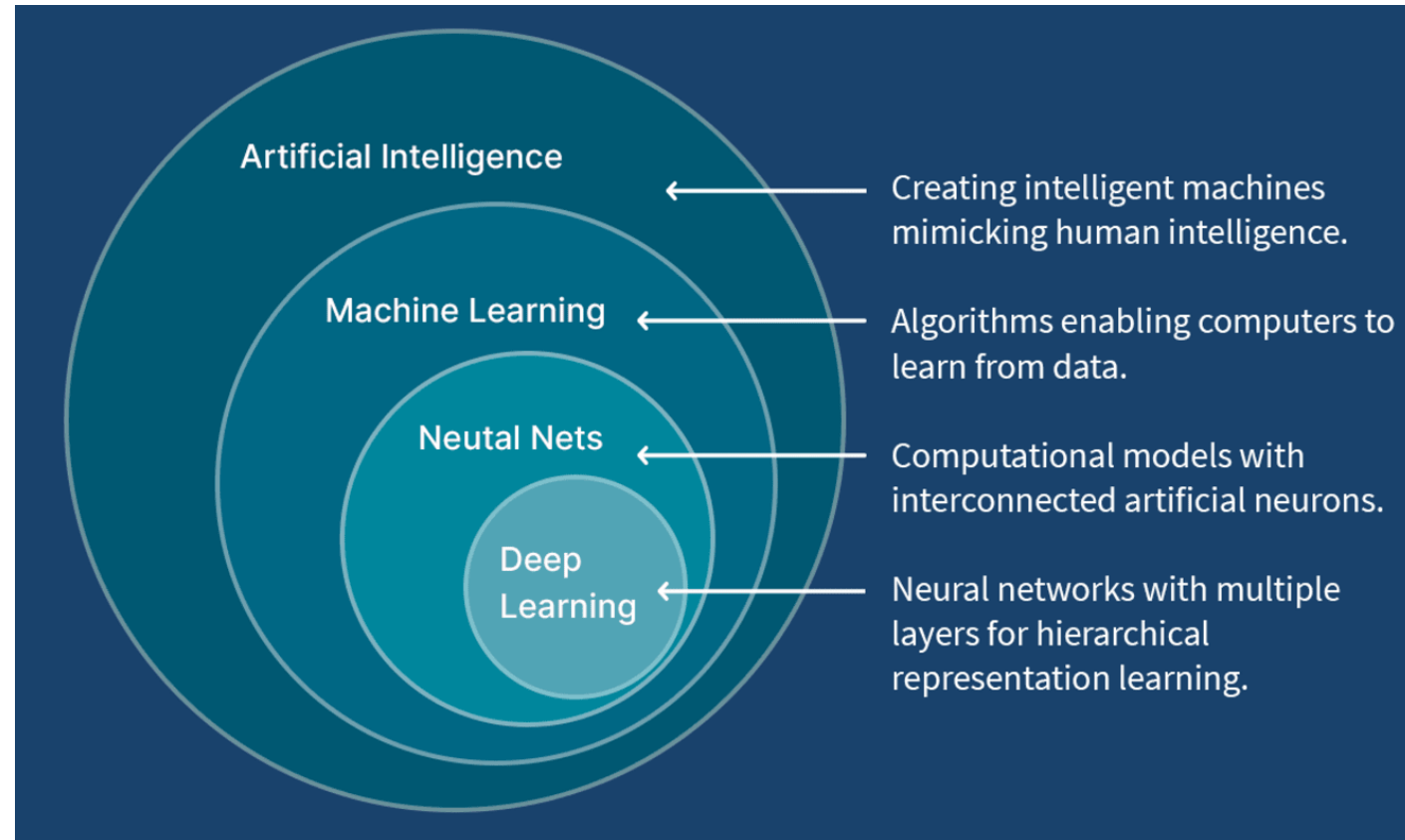
- For case-2 waters, water is dark because of CDOM
- Simpler methods like Blue/Green ratios do not work anymore
- Specific ratios like NDCl may struggle over wider ranges of Chl-a

AOPs [R_{rs}] $\rightarrow$ [? ML ?] $\rightarrow$ IOPs (a,b)



AOPs [R_{rs}] $\rightarrow$ [Yes, ML] $\rightarrow$ IOPs (a,b)

- A method that can handle
 - **Complexity**
 - **Non-linearity**
 - **More covariables**
 - **Go beyond regression**



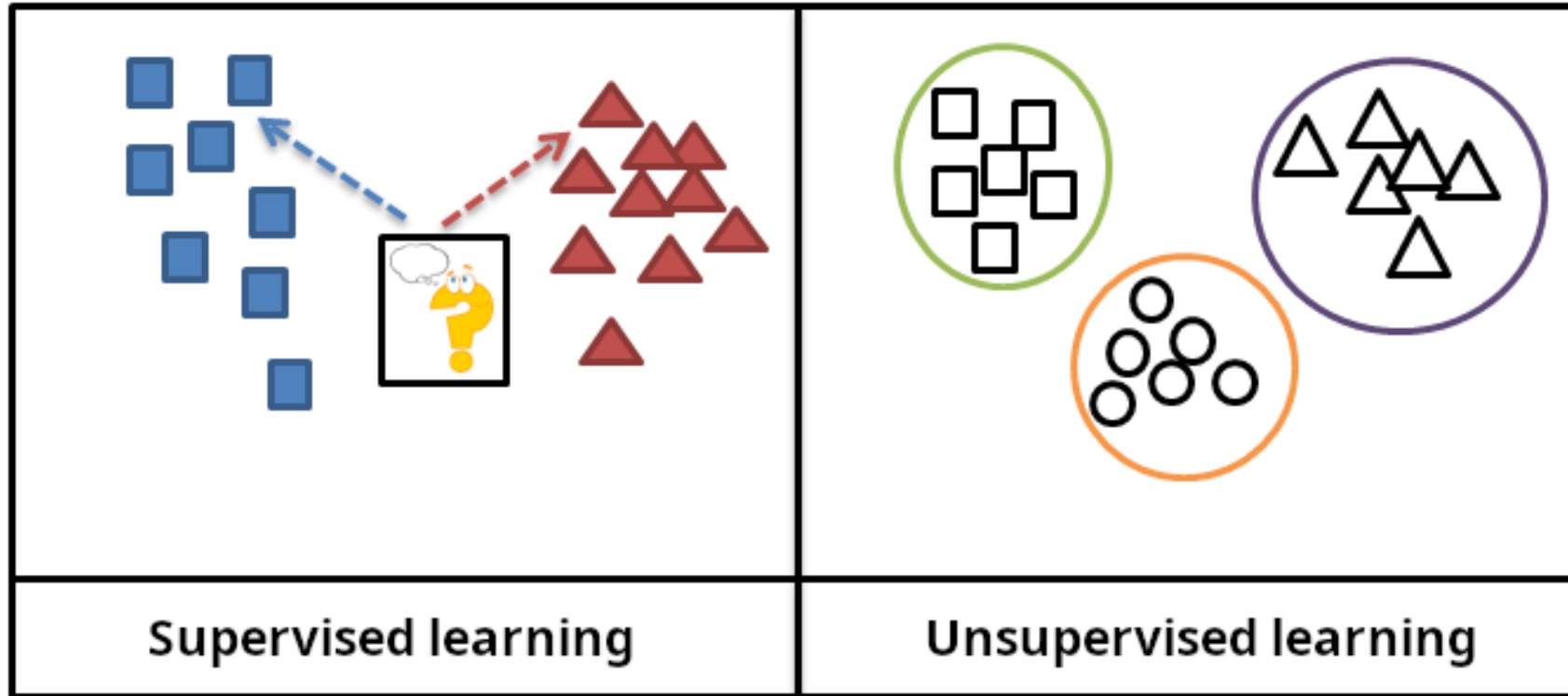
5-min breather

- Reflect on what we talked about
- Does (most of) it make sense?
- Do you have any questions?

An Overview of Machine Learning in Aquatic Remote Sensing

Machine Learning - Types

- **Sup**
- Train
- Analyze

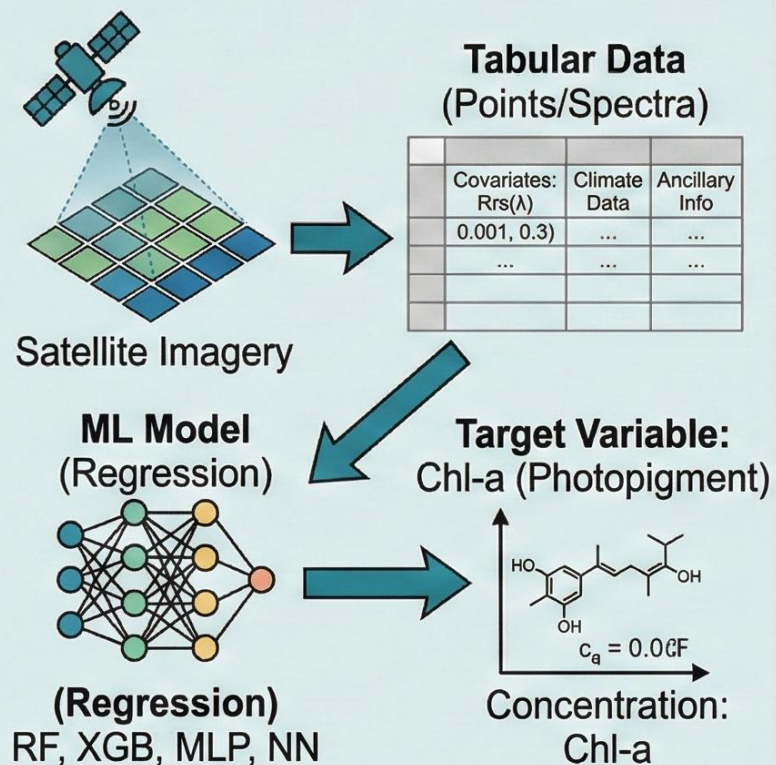


rectly
test,
out

Machine Learning Paradigms in Aquatic Remote Sensing

1. Tabular Machine Learning (Regression-based)

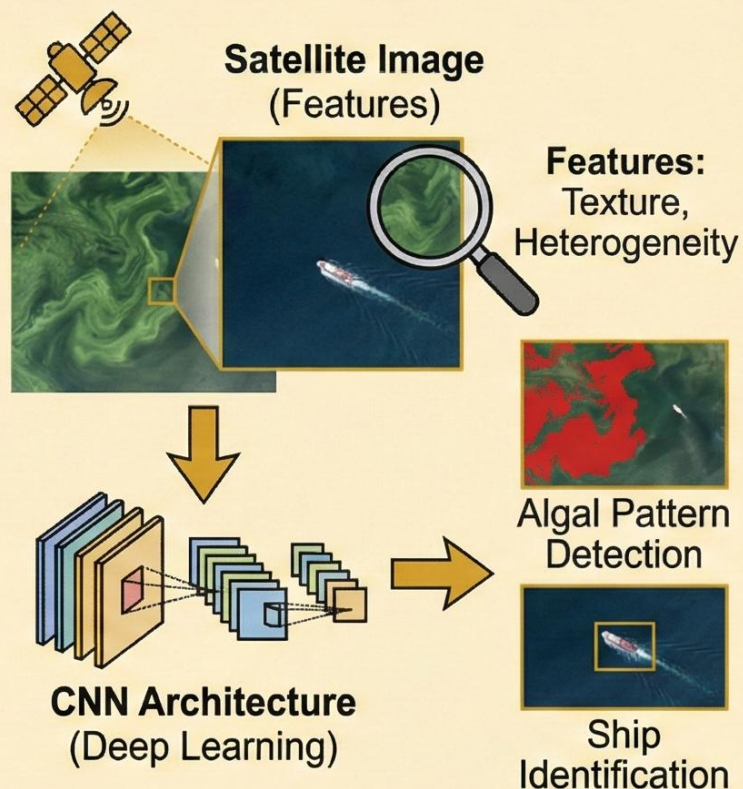
Process Flow & Key Characteristics



Focus: Single Point/Spectra Prediction.
Surrounding pixels less critical.

2. Vision-based ML/AI (Pattern Recognition)

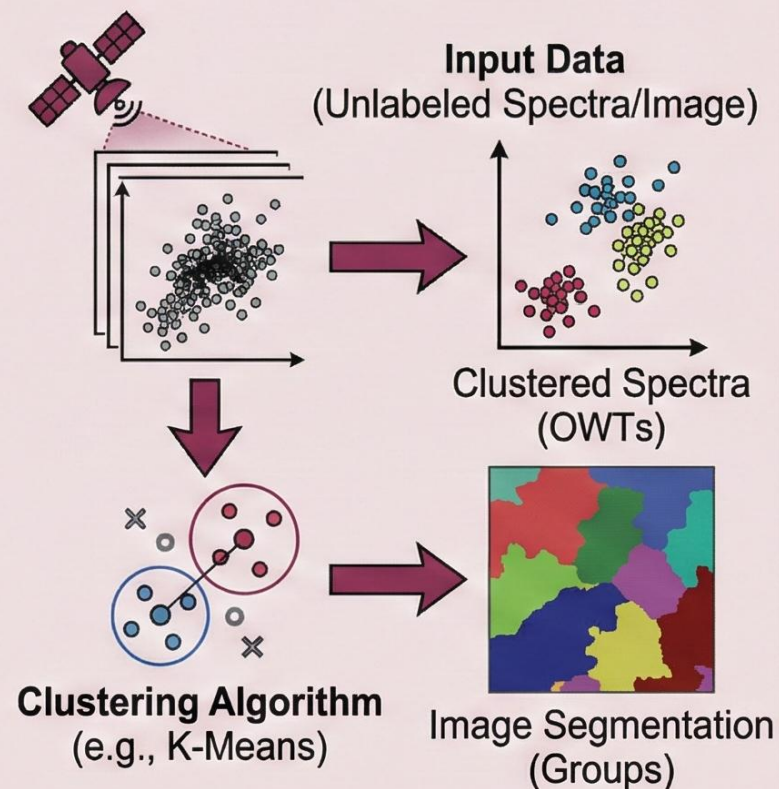
Process Flow & Key Characteristics



Focus: Image Patterns & Spatial Context.
Surrounding pixels matter.

3. Unsupervised Clustering (Grouping & Segmentation)

Process Flow & Key Characteristics

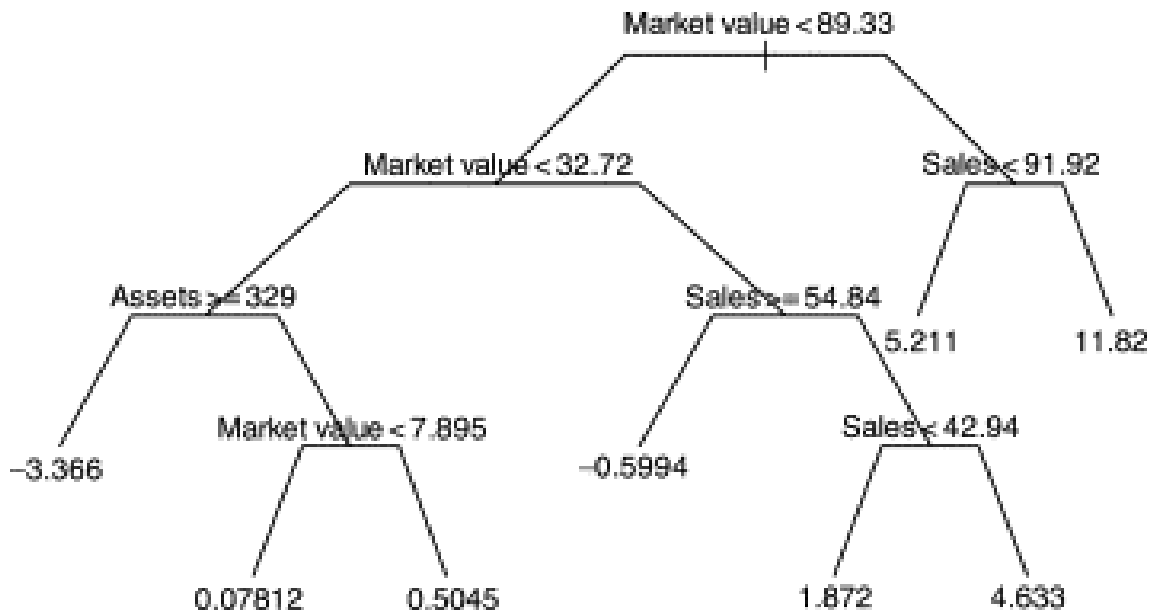


Focus: Similarity & Grouping.
No explicit target variable.

Tabular Machine Learning Algorithms

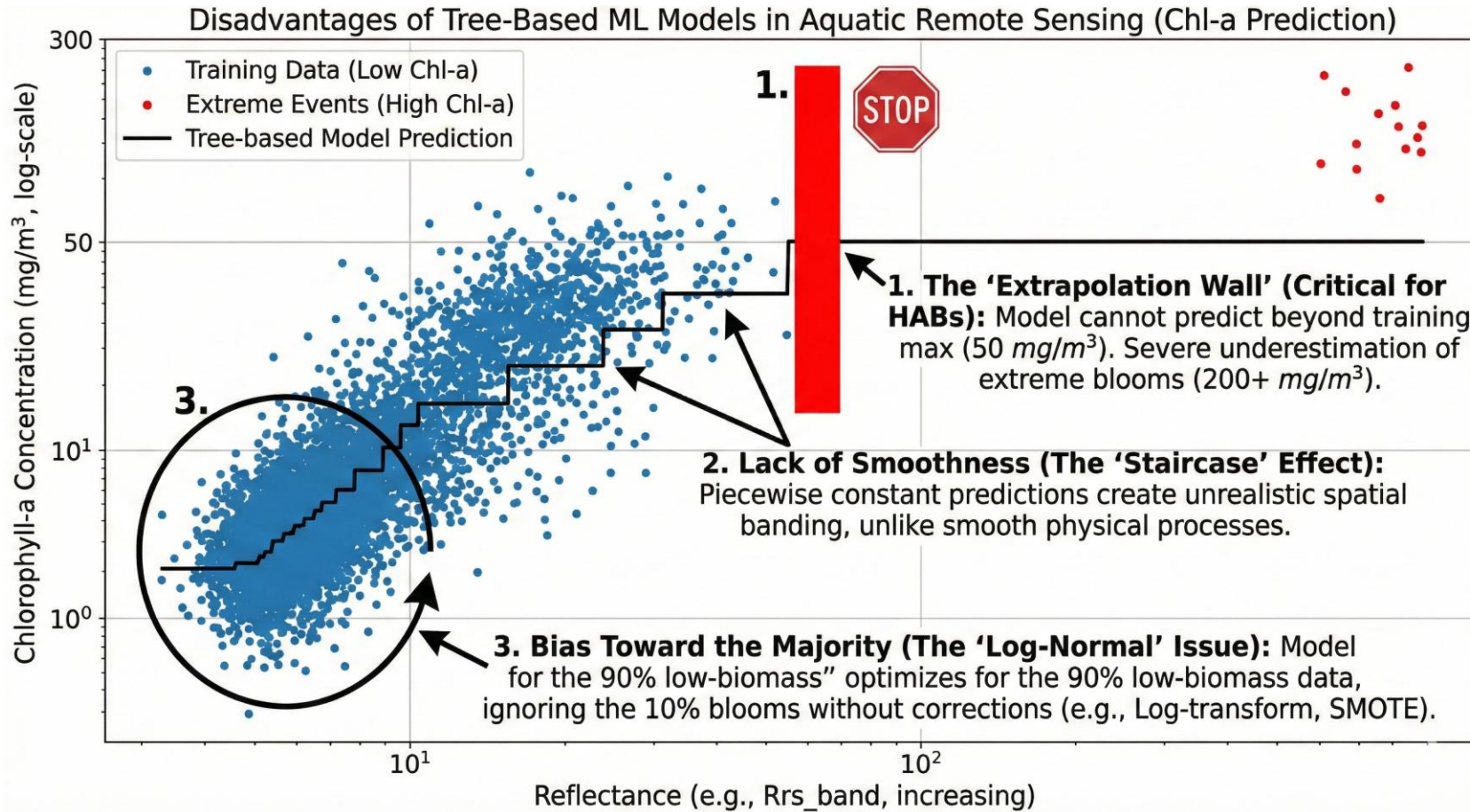
- Main Idea
 - Tabular data – regression / classification
 - One (or more) “target” variables
 - One (or more) “predictor” variables
- Think: regression / linear or multi-linear kind of setup
- Algorithms
 - Tree-based
 - Support Vector Machines
 - Neural Networks

Tree-based ML algorithms



- Advantages
 - Handle Case-2 non-linearity naturally
 - No assumptions about underlying data
 - Excel at separating water-types
 - Built in feature selection
 - Very interpretable

Tree-based ML algorithms



- Disadvantages
 - Very sensitive to range of data – cannot extrapolate (like a look-up table)
 - Bias towards majority of data – goes to distribution of target variables

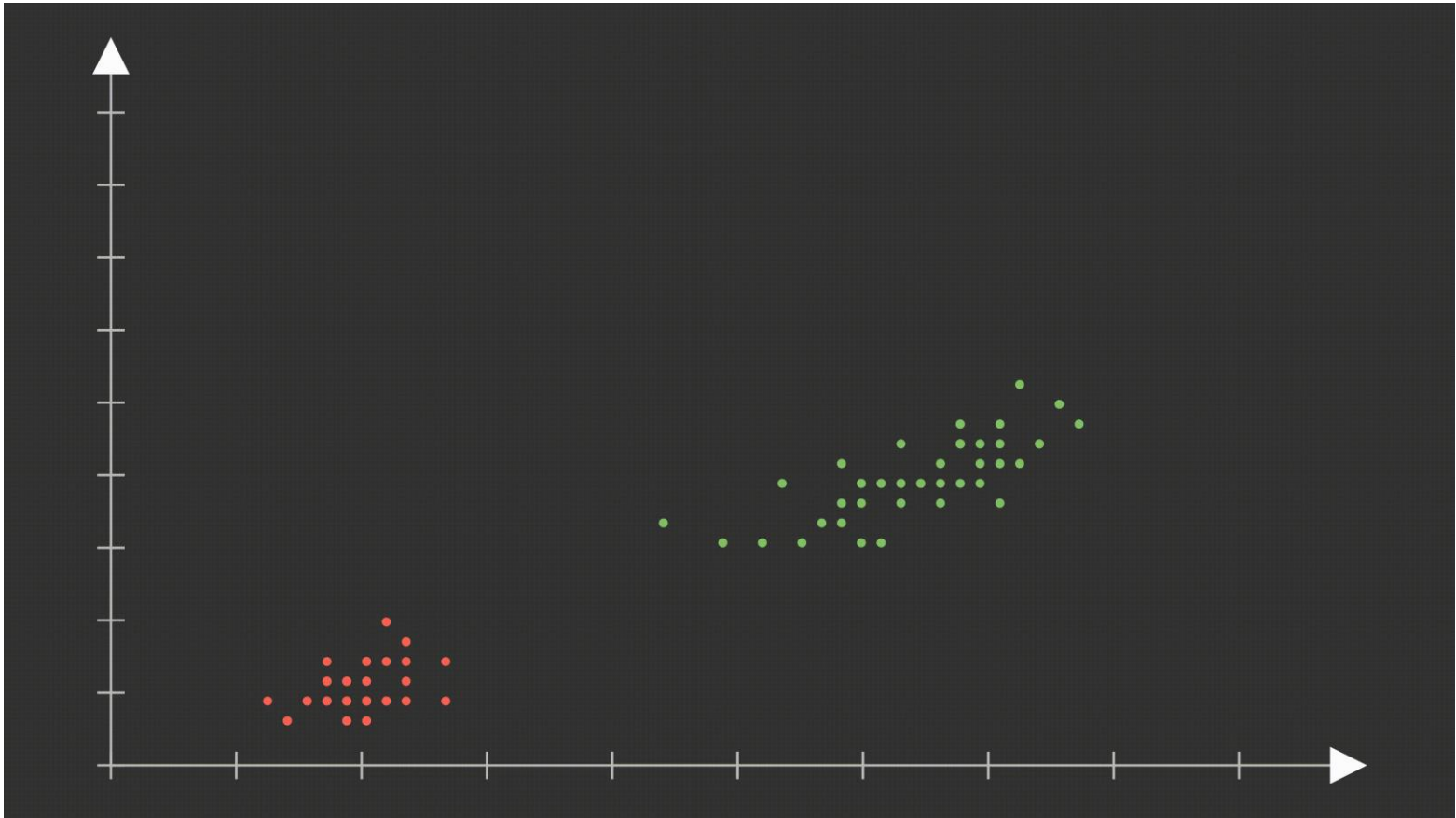
Tree-based ML algorithms

- Decision Trees
- Random Forest
- Extreme Gradient Boosting
- Ensemble Models

“Trees are great interpolators, but terrible extrapolators. If you want to find new patterns *within* your known water types, use Random Forest. If you want to predict a record-breaking event you've never seen before, be very careful.”

Support Vector Machine (SVM)

- Suitable for regression
 - Chl-a concentration
- Suitable for classification
 - OWT
 - LULC



Support Vector Machine (SVM)

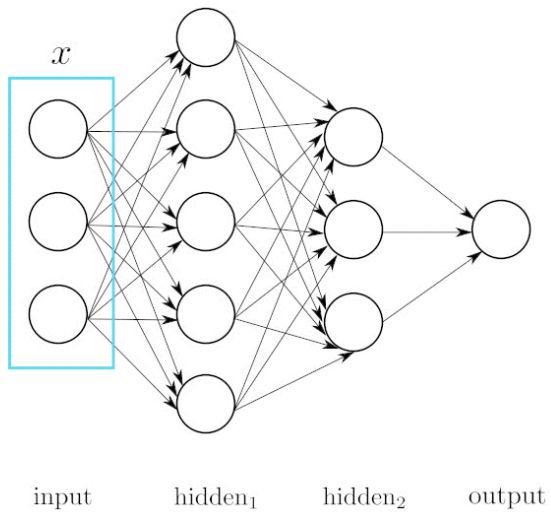
Advantages

- “Small data champion”
- Handles high dimension data well
- Robust to outliers

Disadvantages

- Not scalable
- Not interpretable (black box-y)
- Sensitive to hyperparameters (must tune perfectly to work well)

Neural Networks



Start with the input layer:

- Literal steal from biology – simulation of human neuron
- Inputs: Bands/Rrs, other predictors
- Weights: Strength of each connection (this is what the model *learns*)
- Universal approximator – can learn *any* pattern in nature, no matter how complex
- We went from ML → DL. Important concept in next slide.

ML → DL

- In machine learning, the more you engineer your features, the more you help your model.
 - For eg: 665nm and 708nm are highly correlated to Chl-a
 - High TSM may also be correlated to Chl-a presence
 - You **explicitly** tell your model this by extracting 665, 708 and TSM as features
- In deep learning (Neural Networks), you want your input data to be as raw as possible
 - The NN is a universal estimator – it will learn patterns from the data you feed it
 - If you extract features, you are limiting the “universal space” for it to learn

Neural Networks

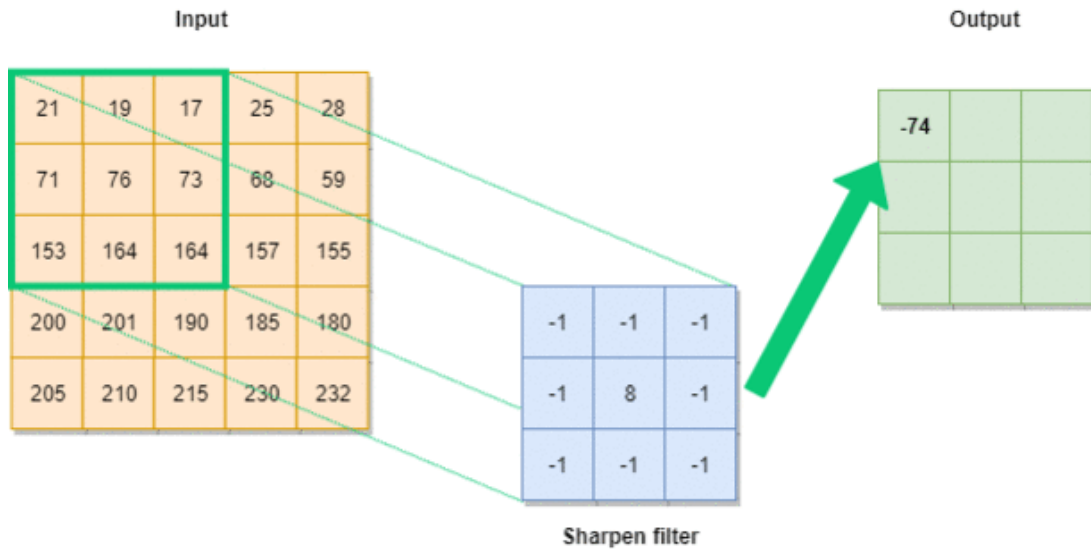
Advantages

- Universal approximation
- Smoothness and physical realism
- Excellent feature extractors – 1D Convolution

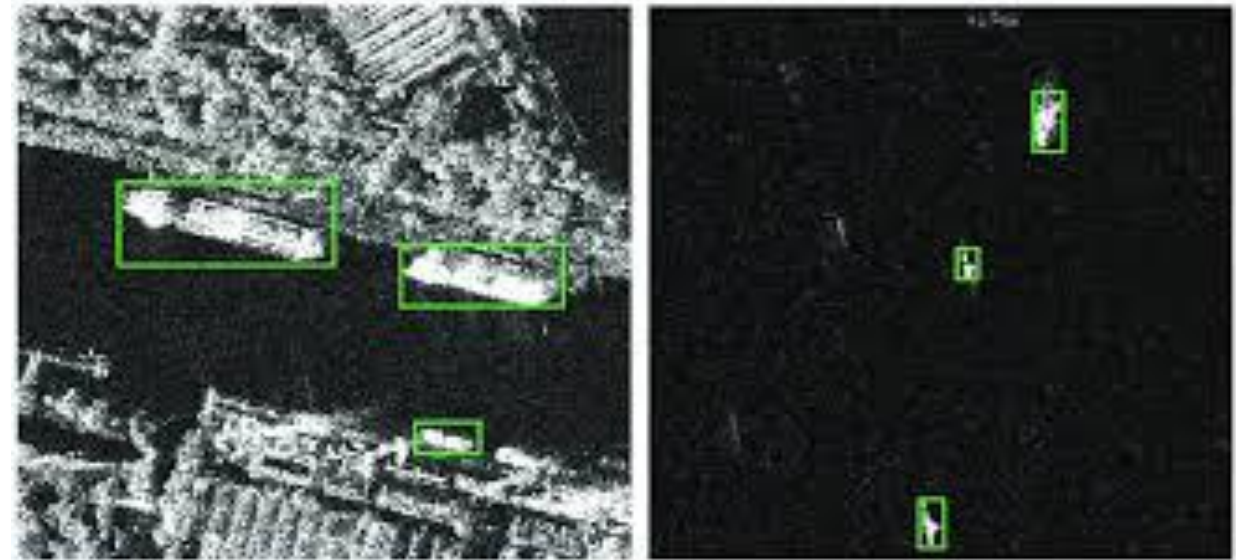
Disadvantages

- Data hungry
- Translucent in explainability – sometimes hard to pinpoint fault
- Extrapolation is without bounds – mathematical shoot off!

Vision-based ML/AI

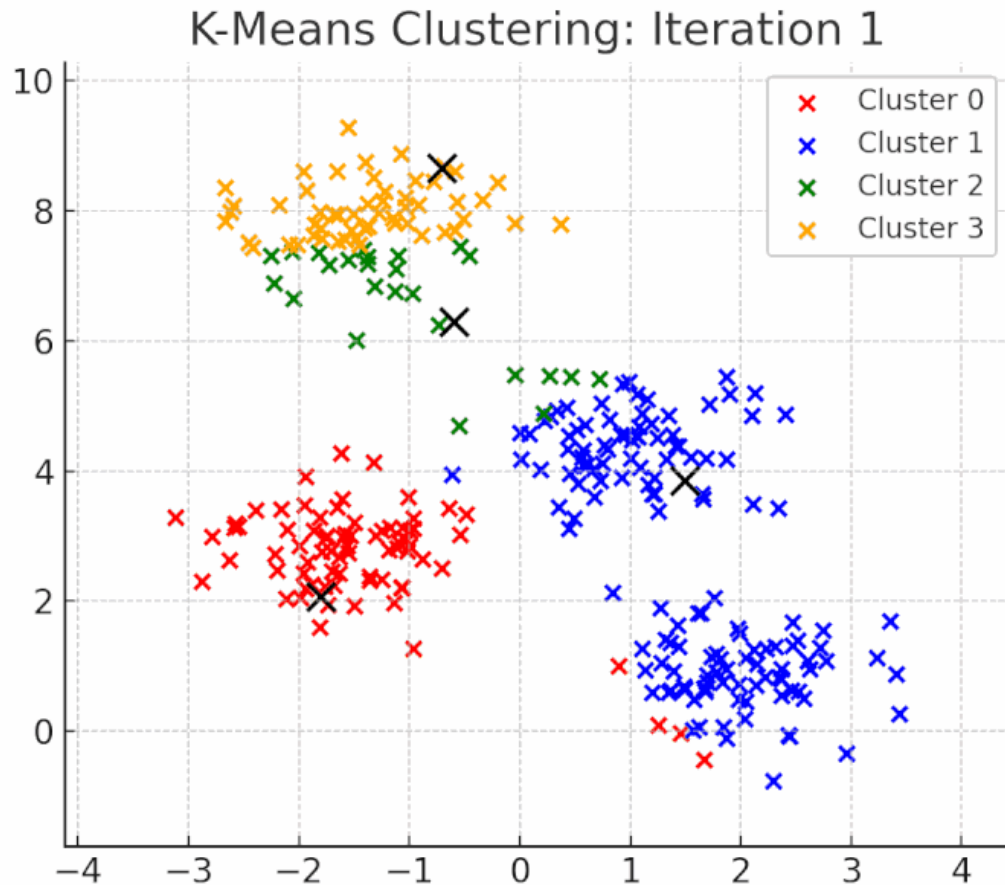


AIGeekProgrammer.com © 2019

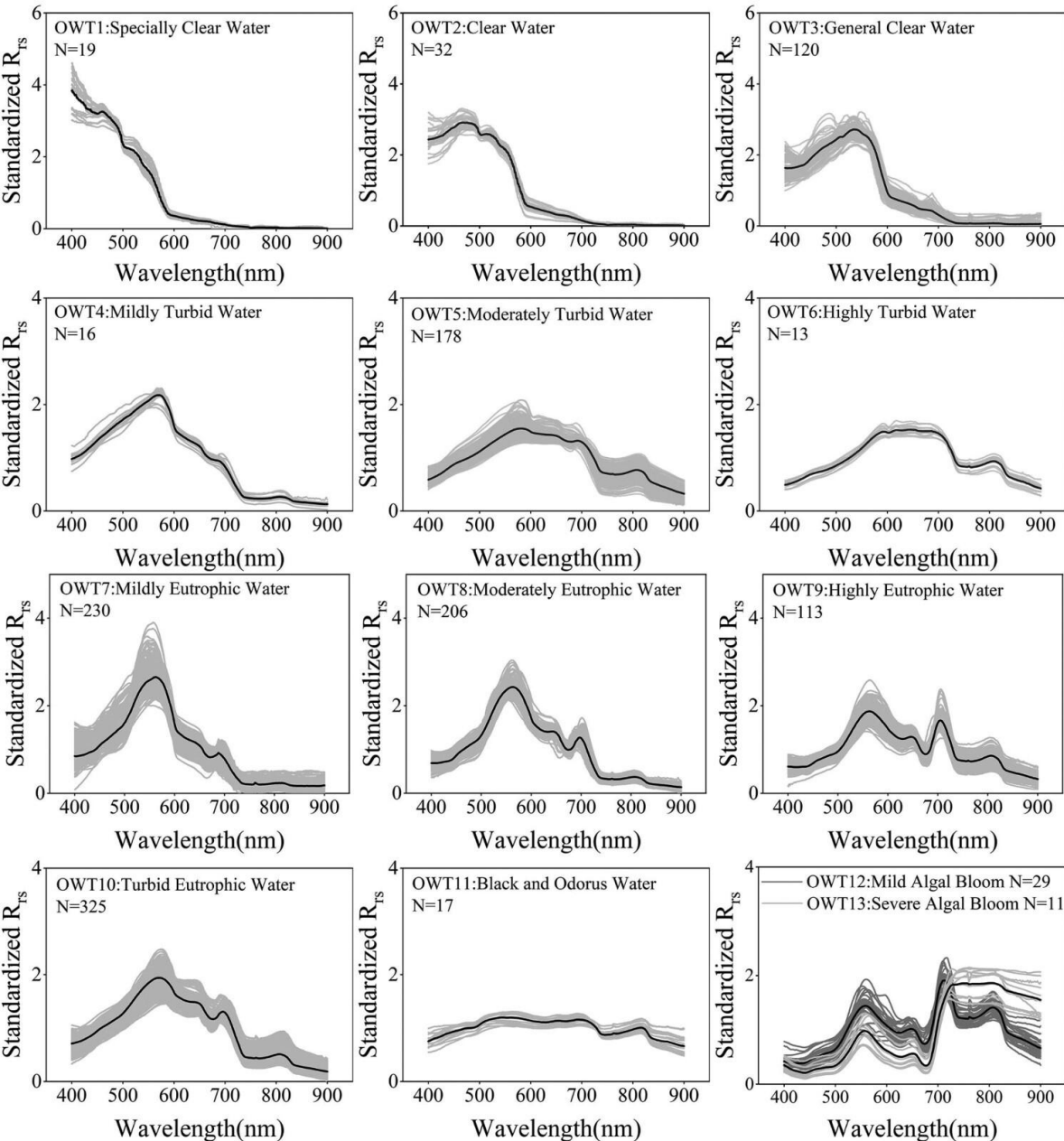


- At image level
- Surrounding pixel matters

Unsupervised Clustering



- Random centroids are initialized
- Points are grouped by nearest centroid
- Centroids are recomputed
- Repeat until clusters are stabilized (centroids don't move significantly)



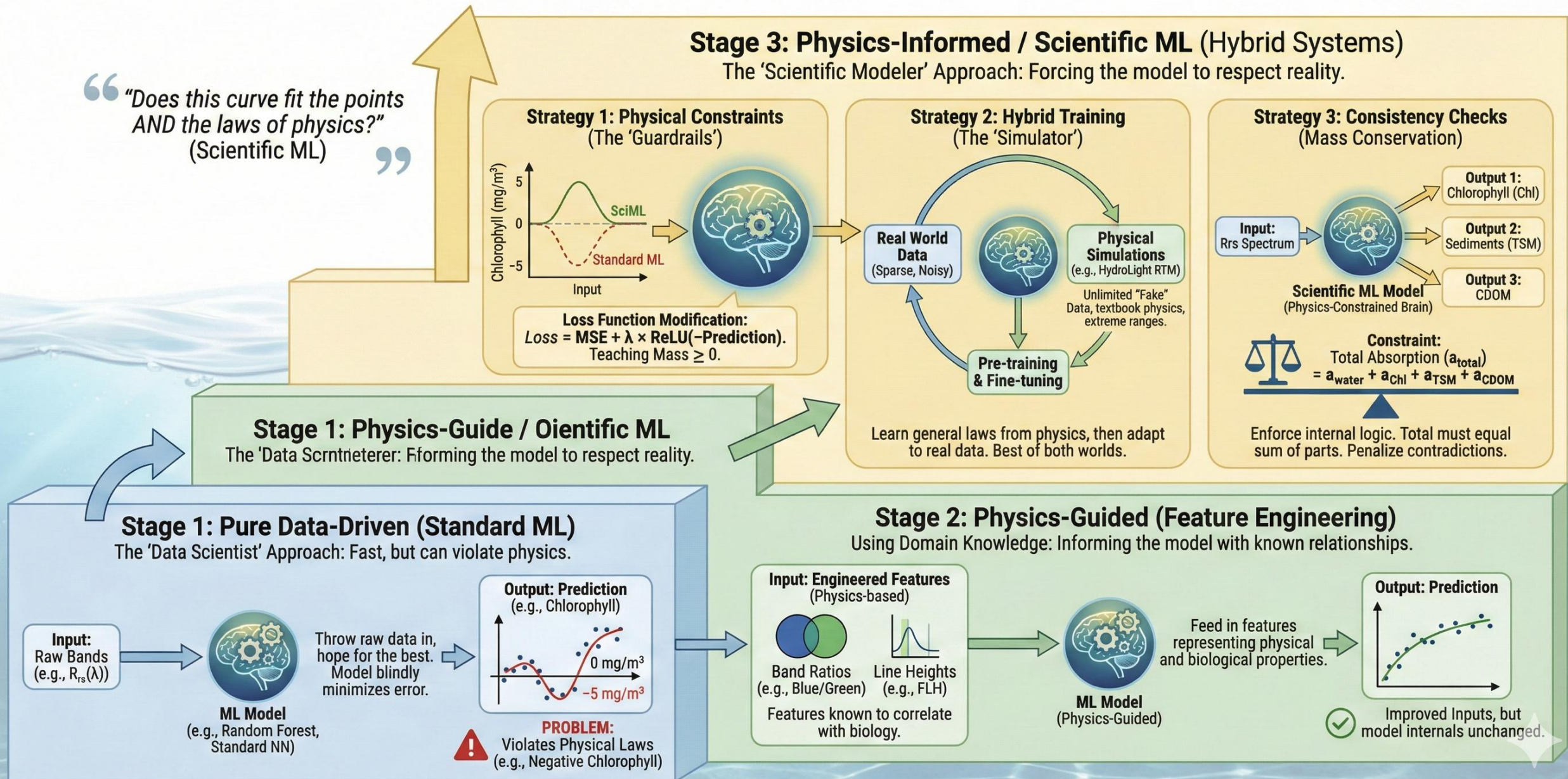
Unsupervised Clustering

Applications

- Optical Water Type Clustering
- Image-level operations such as pattern recognition

The Evolution to Scientific Machine Learning (SciML) in Aquatic Remote Sensing: From Data to Physics

“Does this curve fit the points
AND the laws of physics?”
(Scientific ML)



5-min breather

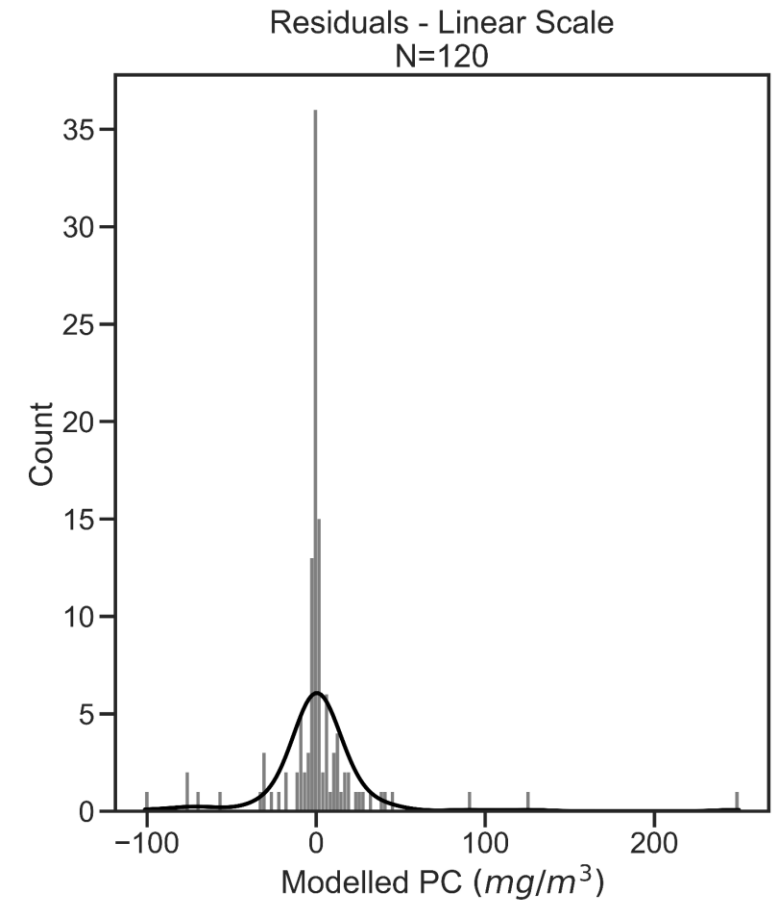
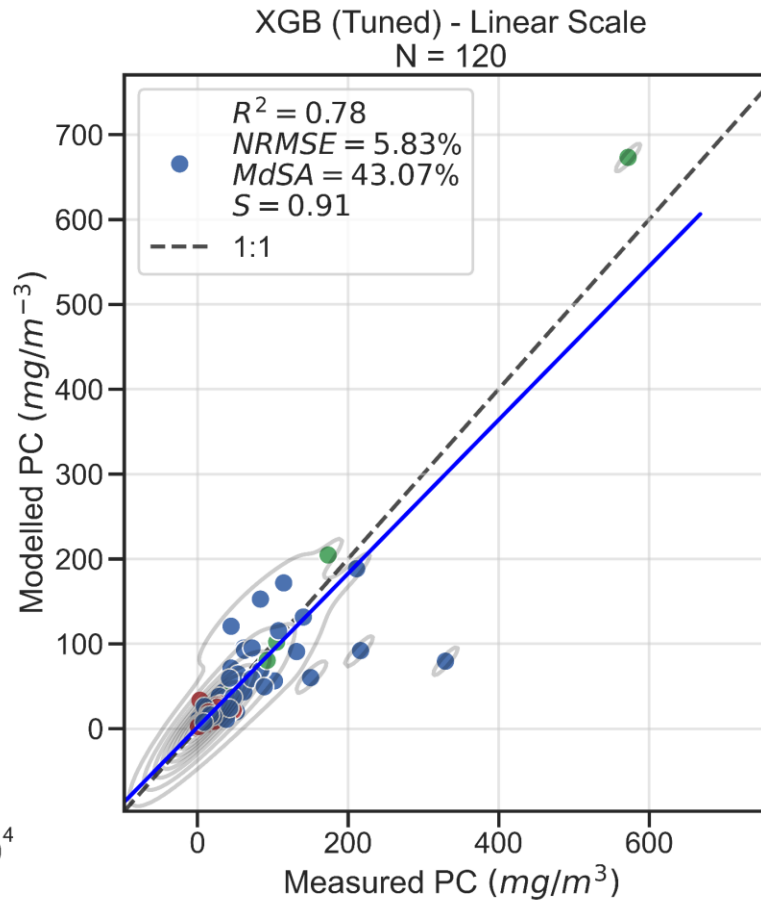
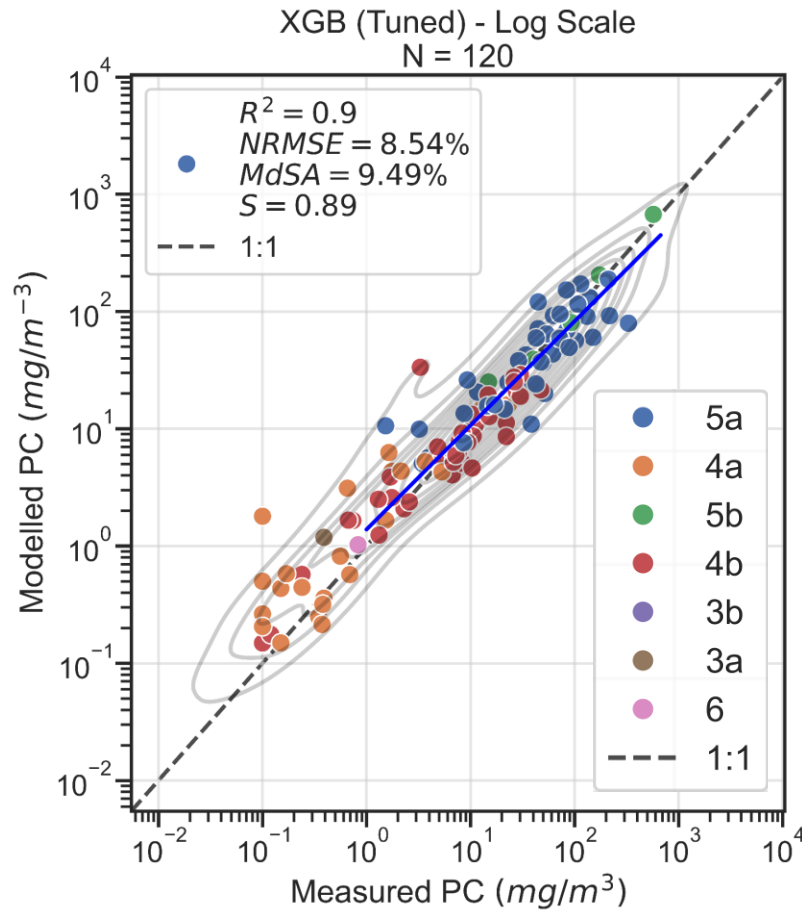
- Reflect on what we talked about
- Does (most of) it make sense?
- Do you have any questions?

Visualization, Evaluation Metrics: To R^2 and Beyond!

Visualization

- Validation Plot
 - Log scale
 - Linear scale
 - 1-1 line
 - Validation fit
- Histogram of residuals
 - Shows any skew in the model

Always have these three plots!

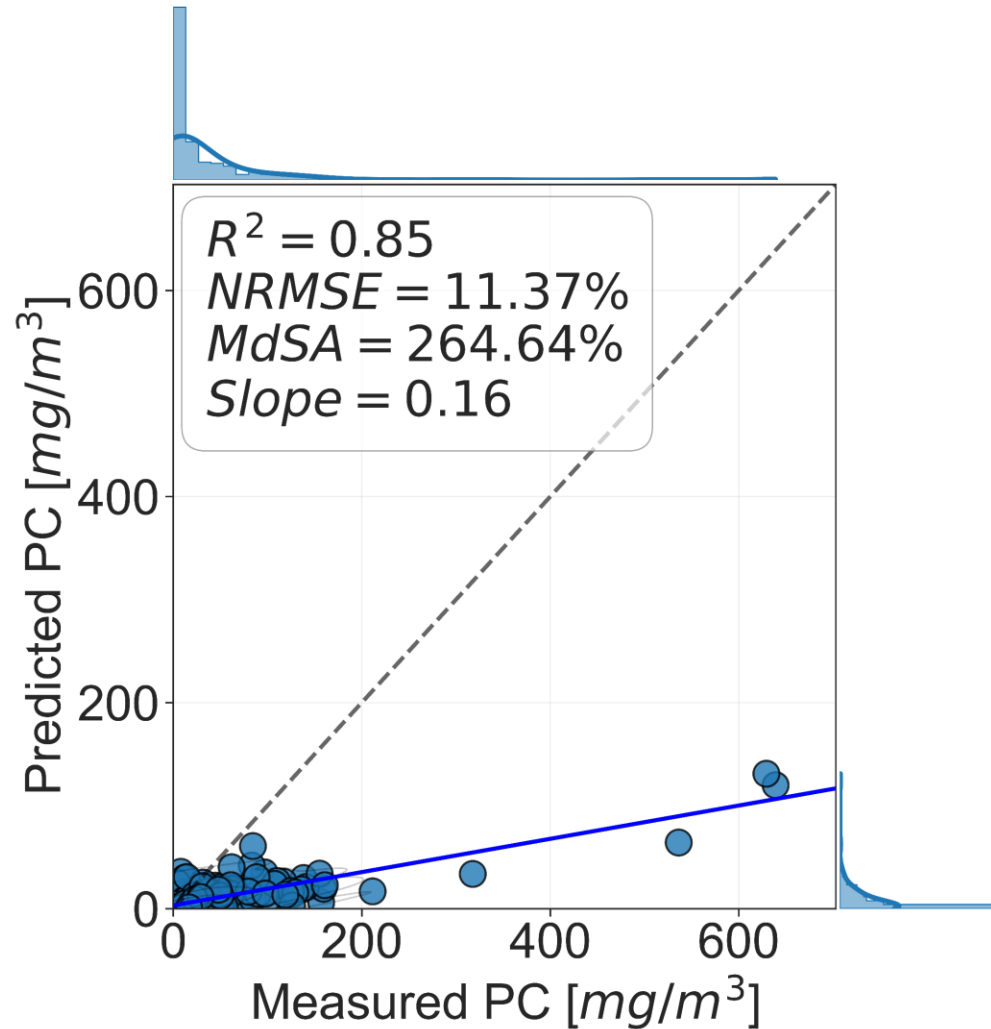


Uncertainty, Metrics and Scale

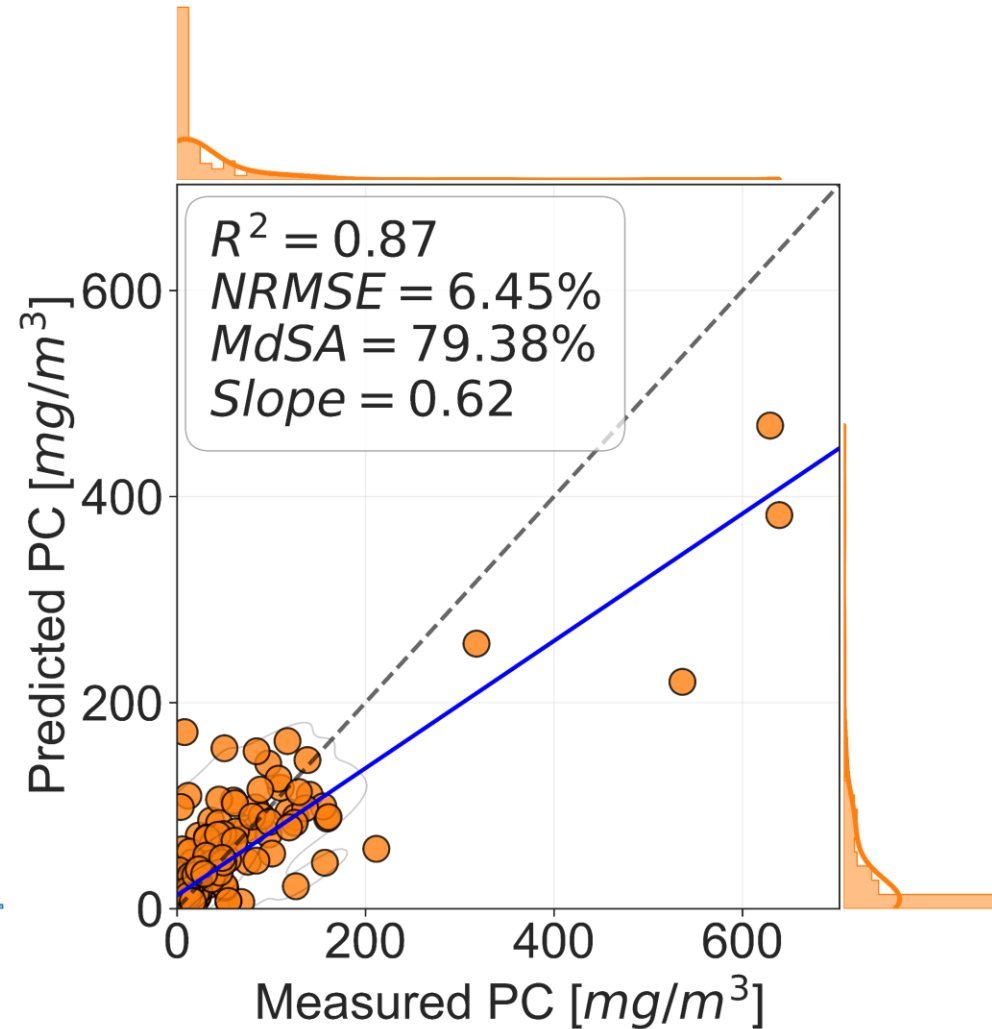
- High R^2 doesn't always mean the best model!
 - It assesses fit, based on data spread but not based on 1-1 line
- Look at RMSE, %NRMSE, MAPE, Bias, Slope, MdSA
- Special emphasis on **scale of data**
- Log scale
 - Okay for visualization
 - Use cautiously for learning
 - Always cross present metrics across log and linear scale for consistency

Why beyond R^2 ?

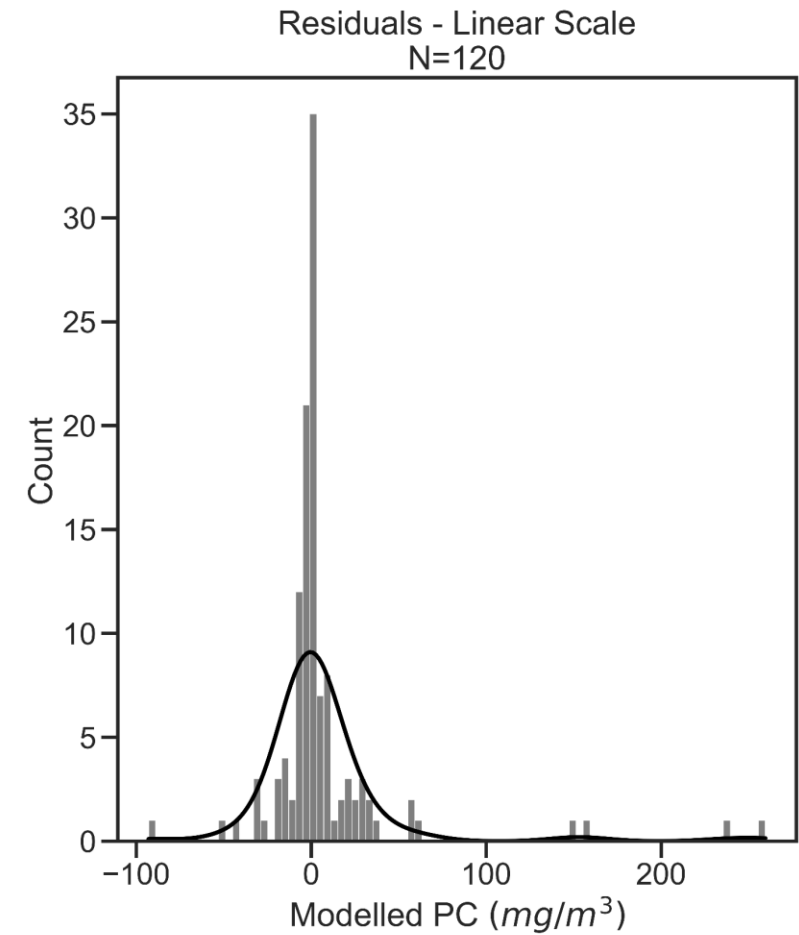
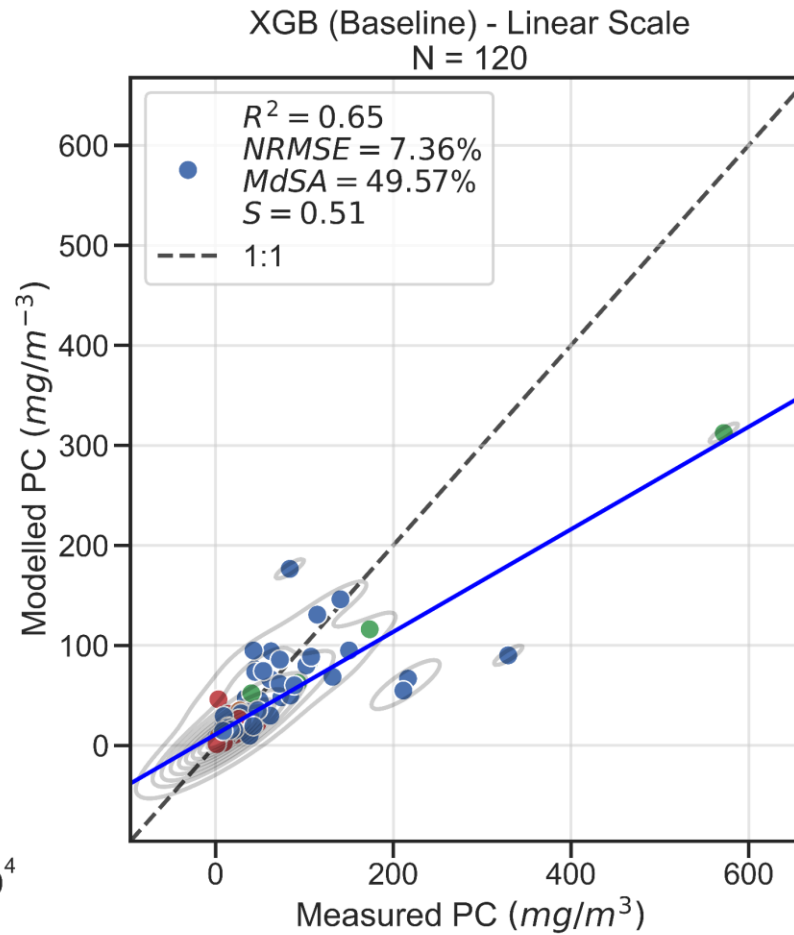
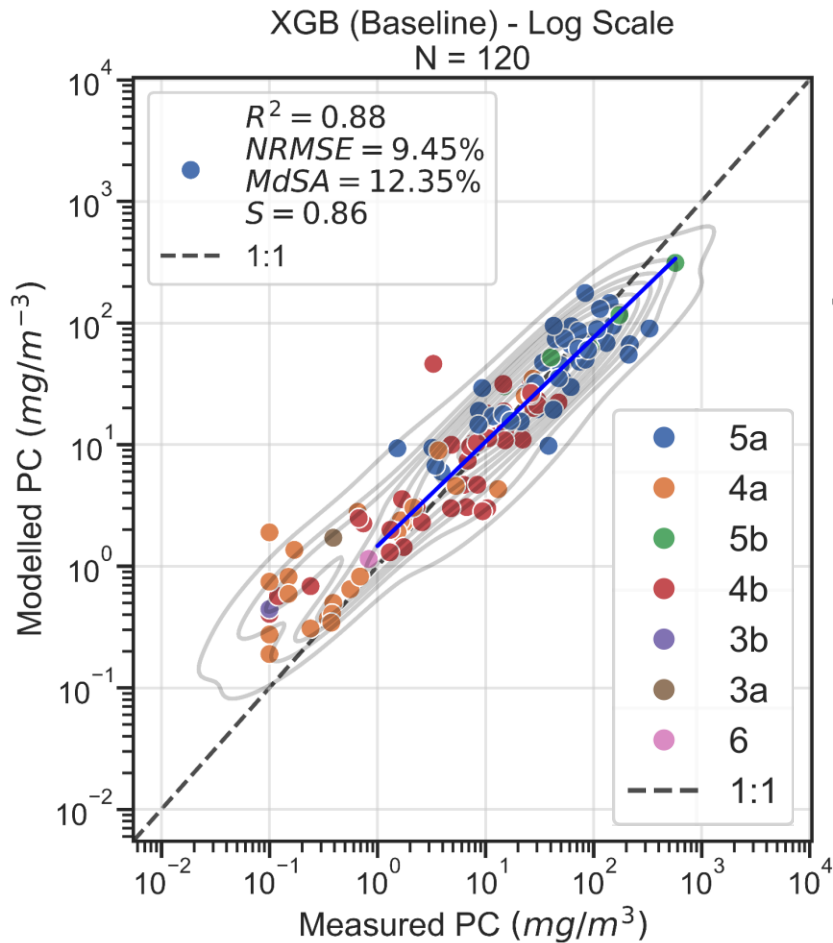
Gaussian NN (η); $N = 224$



Gamma NN (μ); $N = 224$



Why careful with log scale?



5-min breather

- Reflect on what we talked about
- Does (most of) it make sense?
- Do you have any questions?

Ethics/Morals while using ML

“scientists first, data scientists second!”

The Philosophy (Honesty > Perfection)

“*All models are wrong, but some are useful.*” — George Box

The goal of scientific ML is not to sell a perfect R^2 . It is to **honestly quantify reality**.



A model that admits uncertainty is infinitely more valuable than one that confidently guesses wrong. Our job is to report the error bars, not just the answer.

The ML Toolbox (The “Stack”)

THE LANGUAGE (Foundation)



Dominant
force in ML



Strong for
statistics

THE PLATFORMS (Cloud Computing)



Microsoft
Planetary
Computer



THE ML ENGINES (The “Brains”)



PyTorch



THE DATA HANDLERS (The “Glue”)



Pandas



Rasterio



KEY BULLET POINTS



The Language:

Python is the industry standard for ML;
R remains strong for statistical analysis.



The ‘Brains’ (ML Libraries):

Scikit-learn: “Swiss Army Knife” for
tabular ML (RF, SVM, Regressions).

XGBoost / LightGBM: High-performance
Gradient Boosting (Kaggle winners).

PyTorch / TensorFlow: Deep Learning
frameworks for complex neural
networks.



The ‘Glue’ (Geospatial Libraries):

Xarray: “Pandas for Satellite Data”
(N-dimensional NetCDF).

Rasterio: Reading/writing GeoTIFFs.



The Cloud:

Google Earth Engine (GEE): Good for
rapid prototyping and simple Random
Forests.

Microsoft Planetary Computer:
Massive data access coupled with
Python computing environments.

LLMs: Friend or Foe?

- LLMs can sound very “*convincing*” – be careful learning a new concept. Always go to the source (literature or documentation)
- Hallucinations – LLMs can often spit out code or a conceptual explanation that may not be true. For code, you will discover it when you run it → Error. No way for concept?
- Rule of thumb: “Never run code you cannot explain”!

LLMs – Friend or Foe?

The 'Safe Zone' (Friend - Junior Coder)

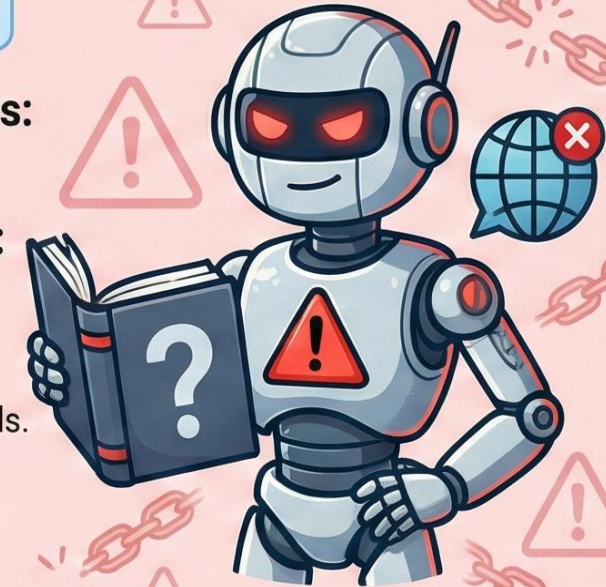


The Motto: Use it as a Junior Coder, not a Senior Scientist.

- ✓ **Syntax & Debugging:**
Fixing code errors.
- ✓ **Boilerplate Code:**
Generating repetitive structures.
- ✓ **Explaining Error Messages:**
Translating cryptic logs.
- ✓ **Documentation Lookup:**
Finding standard library functions.

The 'Danger Zone' (Foe – Senior Scientist)

- ⚠ **Scientific Theory & Citations:**
Inventing facts or sources.
- ⚠ **Physical Constants & Units:**
Using incorrect values.
- ⚠ **Algorithm Selection:**
Recommending inappropriate models.
- ⚠ **Data Interpretation:**
Misunderstanding domain-specific nuance.



The 'Safety Protocol' (Key Bullet Points)



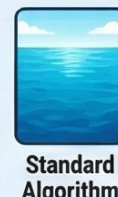
The 'Hallucination' Trap

Confident invention of non-existent libraries or functions. Always check official docs.

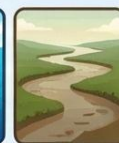


Physics-Blindness

Suggests normalizations that destroy physical signals (e.g., negative R_{rs}). Verify physics.



Standard Algorithm



Custom Needed

The 'OC4' Default

Defaults to standard algorithms (like OC4) unfit for complex waters. Be specific.



The Golden Rule

Never run code you cannot explain. Ask the LLM to explain line-by-line before execution.

Thank you for listening!

- We will discuss more concepts with practical implementations in the next session
- Please feel free to ask any questions/comments/feedback you may have!
- You can reach me at:
chintanmaniyar@gmail.com
- Or LinkedIn:

